



Spiking Neural Network Architecture Optimization on FPGA

Made by: Neural Network Team



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1 Critical Path Calculation

1.1 Hardware Delay Metrics & Assumptions

To determine the maximum operating frequency and critical path across the SNN architecture, the delay for each pipeline stage is measured in uniform "Delays," where the latency of a multiplier is equivalent to the latency of an adder (1 Multiplier Delay = 1 Adder Delay).

The pipeline is explicitly broken down into the following functional delay blocks across the stages:

- Conv Phase: Consists of spatial window multiplication and addition.
- Channel Adder Tree: $\lceil \log_2(\text{Input Channels}) \rceil$ delays (Applied in deeper stages to sum across input feature maps).
- Conv Bias & ReLU: 1 delay.
- BN & MAX Pooling (Patch_Pool): 4 delays (Composed of 1 Multiplier + 3 Adders).
- SNN (LIF): 3 delays (The multiplier is excluded from the critical path).

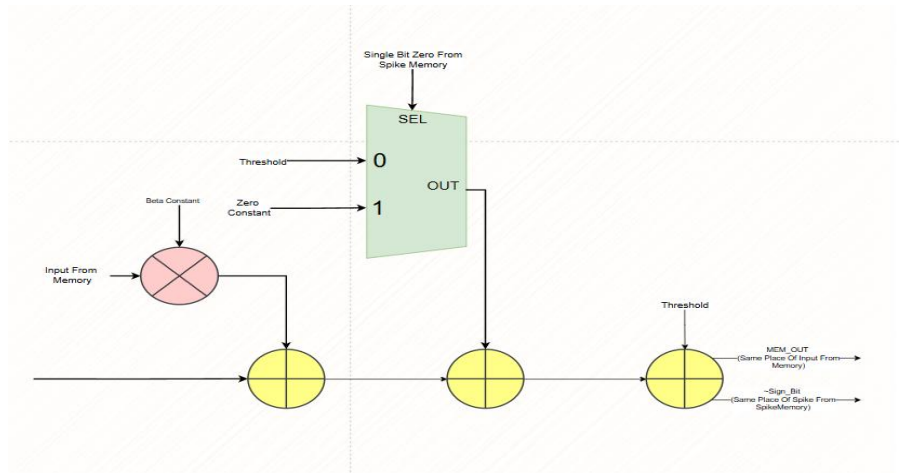


Figure 1: LIF Architecture

1.2 Pipeline Stage Critical Paths

Stage 1 Critical Path

- Conv Block: 6 delays (Composed of 1 Multiplier + 5 Adders processing the initial input)
- Bias & ReLU: 1 delay
- BN & Patch_Pool: 4 delays (1 Multiplier + 3 Adders)
- SNN (LIF): 3 delays
- Total Stage 1 Critical Path: $6 + 1 + 4 + 3 = 14$ delays



Stage 2 Critical Path

- Conv Spatial Window: 5 delays (*1 Multiplier + 4 Adders for 3×3 window*)
- Channel Adder Tree: 5 delays (*Processing 32 input channels from Stage 1*)
- Bias & ReLU: 1 delay
- BN & Patch_Pool: 4 delays (*1 Multiplier + 3 Adders*)
- SNN (LIF): 3 delays
- Total Stage 2 Critical Path: $5 + 5 + 1 + 4 + 3 = 18$ delays

Stage 3 Critical Path

- Conv Spatial Window: 5 delays (*1 Multiplier + 4 Adders for 3×3 window*)
- Channel Adder Tree: 6 delays (*Processing 64 input channels from Stage 2*)
- Bias & ReLU: 1 delay
- BN & Patch_Pool: 4 delays (*1 Multiplier + 3 Adders*)
- SNN (LIF): 3 delays
- Total Stage 3 Critical Path: $5 + 6 + 1 + 4 + 3 = 19$ delays

Summary Note: The overall critical path of the network is bounded by the final computational stage (Stage 3) at a maximum of **19 delays**.

2 Hardware Resource Allocation & Cycle Documentation

2.1 Global Hardware Constraints

The total available computational footprint is capped at 4,638 DSP Blocks. This fixed hardware budget is allocated strictly as:

Maximum Multipliers per Stage: 2,314

Maximum Adders per Stage: 2,324

This maximum resource boundary is applied uniformly across all stages to determine the parallel pixel processing throughput (how many pixels the architecture can process simultaneously per clock cycle).



2.2 SNN Stages Resource and Cycle Summary

The following table summarizes the component-level resource consumption per pixel, alongside the resulting spatial throughput and cycle latency for each structural stage.

Stage	Channels	Conv Resources (Mult / Add)	BN-Pool & SNN Resources (Mult / Add)	Total Resources per Pixel (Mult / Add)	Processed Pixels per Cycle	CycleTotal Cycles
Stage 1	32	100/100	5/10	105/110	20	160
Stage 2	64	1152/1152	5/10	1157/1162	2	512
Stage 3	128	2304/2304	5/10	2309/2314	1	128

Total Pipeline Cycles: $160 + 512 + 128 = 800$ cycles per full output pixel block.

3 Total Processing Time (Latency) Calculation

To determine the total real-world latency required to process a full batch of 16 image frames, we calculate the total pipeline cycles across the defined spatial resolution and divide it by the hardware's operating frequency.

System Parameters:

Cycles per output pixel block: 800 cycles

Spatial Resolution: 32x32 pixels

Number of Frames: 16 frames

Hardware Operating Frequency: 40 MHz (40×10^6 cycles per second)

Time Calculation:

Pixels per frame: $32 \times 32 = 1,024$ pixels

Total cycles required: $800 \text{ cycles/pixel} \times 1,024 \text{ pixels/frame} \times 16 \text{ frames} = 13,107,200 \text{ cycles}$

Total Time in Seconds: $\frac{13,107,200 \text{ cycles}}{40,000,000 \text{ Hz}} = 0.32768 \text{ seconds}$